Introduction to Einstein's General Relativity

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Abstract

These are Lecture Notes for PHYS 413/513 at Rice University. At the moment, they constitute a work in progress. Please report any typos/errors to mustafa.a.amin@rice.edu.

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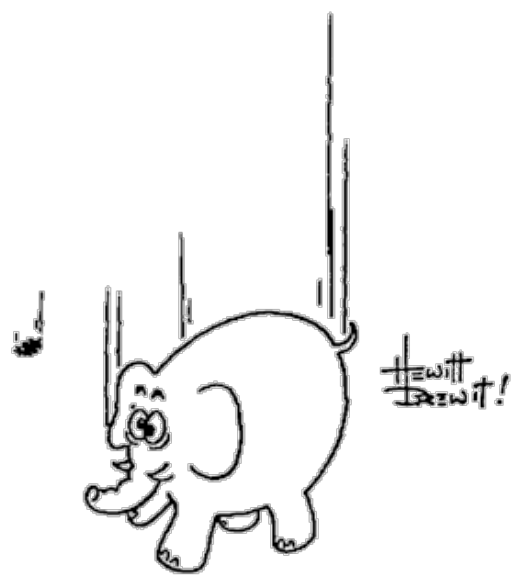

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Preface

These are notes for a one semester course, PHYS 413/513, at Rice University. The course is meant for students (undergraduate and graduate) who have little or no prior exposure to Einstein's General Relativity. However, the students are expected to have a solid foundation in classical mechanics (with Lagrangians), multivariable calculus, differential equations, and linear algebra. Familiarity with Special Relativity and Electrodynamics will be very helpful too.
Students: Do the recommended reading before class!

Acknowledgements

The primary textbook for the course is James Hartle's "Gravity". The notes rely partly on the textbook for organization, and are very much a work in progress. Don't expect complete notes at the moment. I have added my own examples, explanations and visualizations. where I thought I had something to say which can help students further. I will also supplement these notes with exercises to be done as you read through. This will all happen in due time.

I learnt General Relativity first by reading Sean Carroll's more mathematical textbook, which I thoroughly enjoyed. I was enamoured by the elegant differential geometry at the time, but missed some of the relevant physics in that first pass. Hartle's book helped address that for me. I have also benefited from Misner, Thorne and Wheeler's "Gravitation", Weinberg's presentation in "Gravitation and Cosmology" and Wald's book "General Relativity". More recently, I have enjoyed learning from the Special and General Relativity sections of Thorne and Blandford's "Modern Classical Physics".

I owe an immense gratitude to my teachers – Prof. Eva Silverstein who first asked me to learn GR during a "rotation" during the first year of graduate school, Prof. Renata Kallosh for my first course, and Prof. Robert Wagoner for letting me TA for his GR class. This is how I learnt GR during graduate school. Prof. Barbara Shipman at the University of Texas at Arlington taught me multivariable Calculus and Linear Algebra, and then differential Geometry during my undergraduate years, both in class and often on Friday afternoons beyond class. Her enthusiasm for and pedagogy was infectious, and I hope I can follow in her footsteps.

And, importantly, the previous Phys 413/513 students at Rice were critical in making the course what it is now.

CONTENTS

CHAPTER 1

INVITATION

We will explore one of the most impressive achievements of our species. Along the way, I will also tell you how to lose some weight.

Launch a coconut and a tomato with the same velocity from the earth's surface. In absence of air resistance, they will follow the same trajectory![1] This, in itself is astonishing – the trajectory is independent of their mass.

If you were launched with them, they would simply appear to float next to you – gravity seems to have disappeared! More generally, if you were falling freely under gravity, by looking at the tomato and coconut (for a short enough period of time) they would each appear to be moving at a constant velocity, with no gravitational acceleration! [2]

Exercise 1.0.1 Why is it surprising that the coconut and tomato follow the same trajectory? Consider the situation where the gravitational acceleration is rel

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Exercise 1.0.2 Consider a coconut and tomato of mass $m_{\rm c}$ and $m_{\rm t}$ launched from the surface of the earth with velocity $\mathbf{v}_{\rm c}$ and $\mathbf{v}_{\rm t}$. They will follow parabolic trajectories $\boldsymbol{x}_{\rm c}(t)$ and $\boldsymbol{x}_{\rm t}(t)$, with velocity $\mathbf{v}_{\rm c}(t)$ and $\mathbf{v}_{\rm t}(t)$ and acceleration $\mathbf{a}_{\rm c}(t) = \mathbf{a}_{\rm t}(t) = \mathbf{g}$. Write down $\boldsymbol{x}_{\rm c,t}(t)$ and $\mathbf{v}_{\rm c,t}(t)$. Does it depend on the mass $m_{\rm c,t}$? Now imagine at the same time, a sheep named Dolly starts falling freely from some height h. What would the position, velocity and acceleration of the coconut and tomato be as measured in the frame of Dolly? You should find that $\mathbf{g}$ disappears, and Dolly will conclude there is no gravity in the situation. Assume the speeds are small compared to speed of light, escape velocity from earth and heights are small compared to the radius of the earth.

With Einstein as our guide, these simple observations lead us to an astonishing insight into Nature. Gravity is unlike other known forces – it is really about the nature of space and time. Three dimensional space is not a passive arena in which events take place in time, but space and time together – *spacetime* – is an active, dynamical entity. It is a smooth collection of events: a 3+1 dimensional spacetime, whose

[1] Check out a version of this experiment on the moon https://www.youtube.com/watch?v=KDp1tiUsZw8. Also see https://www.youtube.com/watch?v=E43-CfukEgs.

[2] Even if the the coconut and tomato were launched with different individual initial velocities, you would still find that they appear to move at a constant velocity relative to you and each other.

geometry is determined by the mass-energy within it. The mass of the earth "curves" the spacetime around it.

As we will learn, despite the parabolic locus of the trajectory in space, the coconut and tomato are moving in this curved *spacetime* on the "straightest" possible path which is independent of the composition and mass of the fruit. Gravity is a manifestation of this curved spacetime.

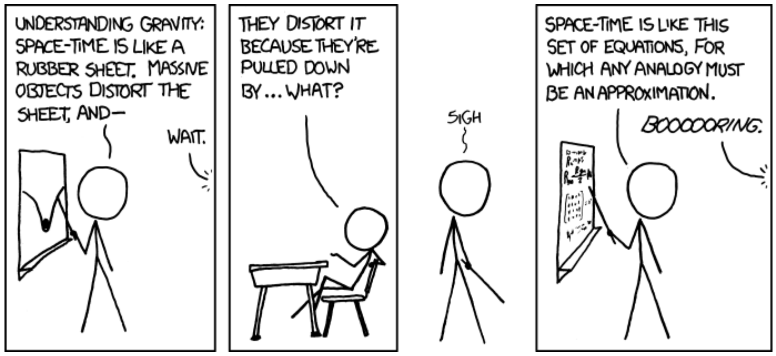

Figure 1.1: Credit: https://xkcd.com

We will spend much of the course developing a quantitative, calculable and testable understanding of the above ideas. This understanding will allow us to appreciate (aspects of) the working of the Global Positioning System (GPS) relevant for your smart phones to help you orient yourself. Moving further out from our abode on our remarkable third rock from the sun, we will learn how the geometric understanding of spacetime explains the puzzling orbit of Mercury around the sun, and the observed deflection of light by the sun[3]. We will be able to describe the formation and environment of spectacular ultra-compact "objects", neutron stars and black holes in our cosmos. We will be able to understand how a dance of their binaries generate ripples in our dynamical spacetime, which have now been detected on our rock.[4] Time-permitting we will touch upon how the evolution of the universe as a whole relies on our understanding of the laws governing a dynamical spacetime.

Our underlying theoretical framework is that of Einstein's General Theory of Relativity (GR). This theory, arguably one of humanity's most elegant and well tested ideas, is built upon Einstein's Special Theory of Relativity. We will review Einstein's Special Theory of Relativity (SR) first – with an emphasis on geometry. The path to GR then is a matter of living with SR "locally" in small enough regions of spacetime, and developing an understanding of how to connect the happenings in many such small regions together. While not obvious to us on earth easily, get in touch with your favorite "freely falling" astronaut in the international space station to understand how gravitational effects (mostly) disappear for them in their "local" laboratory on the space-station.[5] We will compare and contrast our developing understanding of GR with Newtonian ideas of gravity, and see why Newtonian ideas fail to be consistent with SR and more importantly, with observational and experimental tests.

The Newtonian theory of gravity is of course still extremely useful and accurate for most applications on the earth, solar system and in astrophysics more generally. However, Newtonian gravity fails when considering systems with mass M and extent R, such that GM/Rc^2 becomes order unity. Here $c = 3\times10^{10}\mathrm{cm\,s^{-1}}$ is the speed of light in vacuum and $G = 6.67\times10^{-8}\,\mathrm{dyne\,gm^{-2}\,cm^2}$ is Newton's gravitational

[3] Light deflection from the sun was first observed during a total solar eclipse of 1919. For a bit of history and its significance, you might like a talk I gave on this in the context of the solar eclipse that was viewable from the continental US in 2024 https://www.youtube.com/watch?v=D1LGSSoLfNg.

[4] See https://www.nobelprize.org/prizes/physics/2017/press-release/.

[5] In 2024, I read a sign at the Johnson Space Center (inside the mock shuttle), which said astronauts float because of lack of earth's gravity since they are far away from the earth. After this course, you should tell me what you you think about this statement.

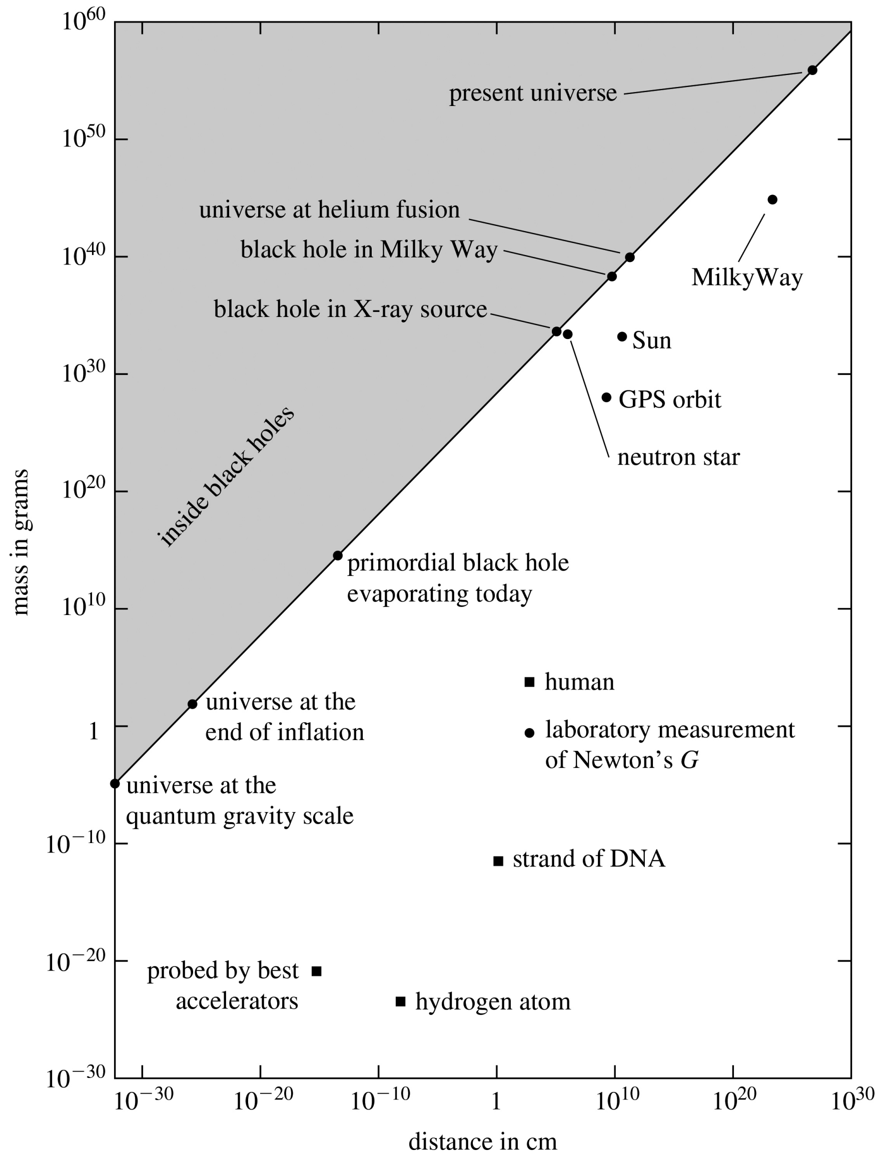

Figure 1.2: General relativity effects are not ignorable near the diagonal line, where $2GM/Rc^2 = 1$. Figure from J. Hartle. To understand the diagonal line, recall that the escape velocity from the surface of a body of mass M and radius R is $v_{\rm esc} = \sqrt{2GM/R}$. When M and R are such that this escape velocity $v_{\rm esc} \to c$, even light cannot escape from its surface (this is the line $M = (c^2/2G)R$). This is the historical definition of a "Blackhole" (Michell & Laplace, 18th century).

constant.[6] Near a Black Hole, $GM/Rc^2 \sim 1$, neutron star $\sim 10^{-1}$, near the sun (say at the orbital radius of mercury) it is $\sim 10^{-8}$, whereas near the earth's surface $GM/Rc^2 \sim 10^{-9}$. Note that near the earth and the sun, the precision of the measurements is sufficiently high that even for such small values of GM/Rc^2, we need to take deviations from Newtonian gravity into account. See Fig. 1.2.

Let us also put general relativity in the context of classical and quantum mechanics. A nice visual way is via the "cube of physics", see Fig. 1.3. You should read this as, for example, when G becomes more relevant, then Newtonian gravity plays a prominent role. We will not have much to say about when

[6] It is very useful to remember that for the sun $GM_\odot/c^2 \sim 1\,\mathrm{km}$, where $M_\odot \sim 3 \times 10^{30}$ kg.

quantum gravity in this course. It becomes important at the Big Bang and near the center of black holes. Notice the physical scale where quantum gravitational effects need to be taken into account. For this course, we will restrict ourselves to classical gravity.

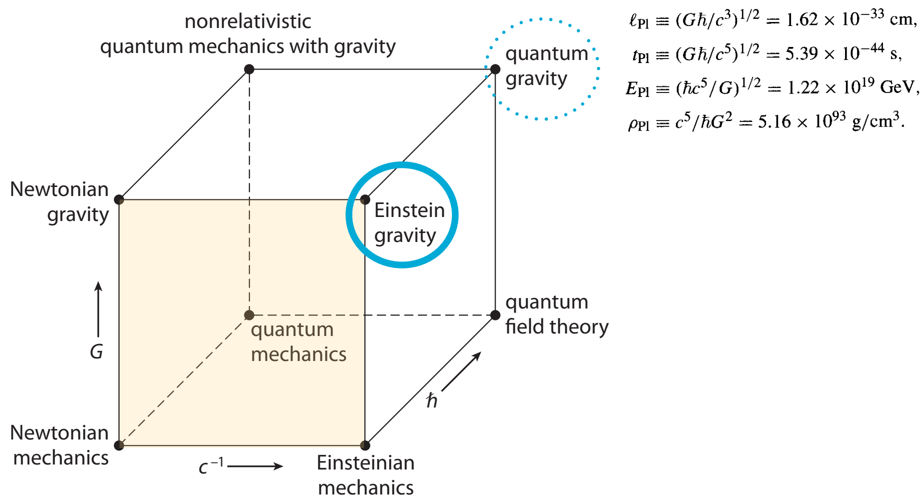

Figure 1.3: Cube of Physics, adapted from A. Zee.